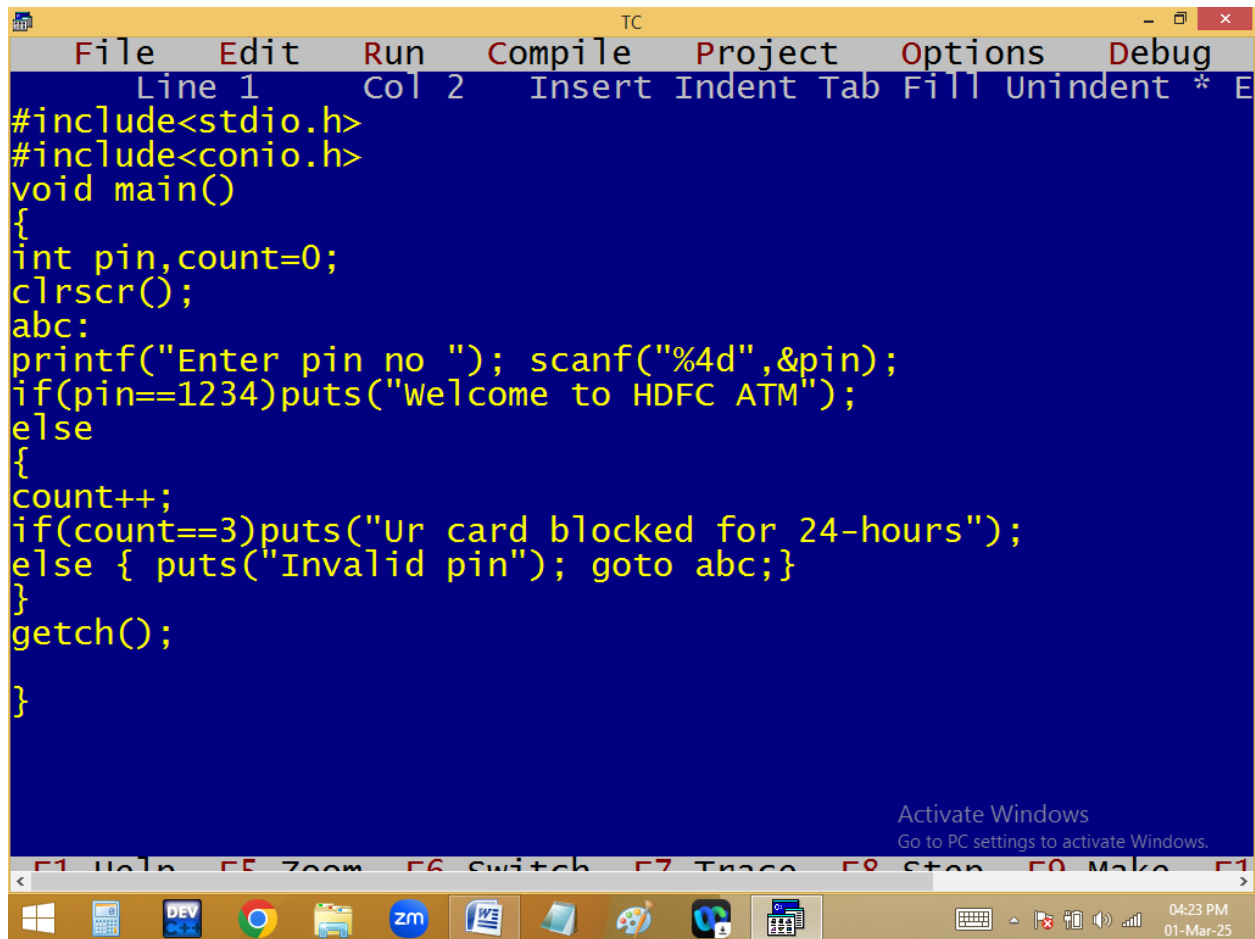


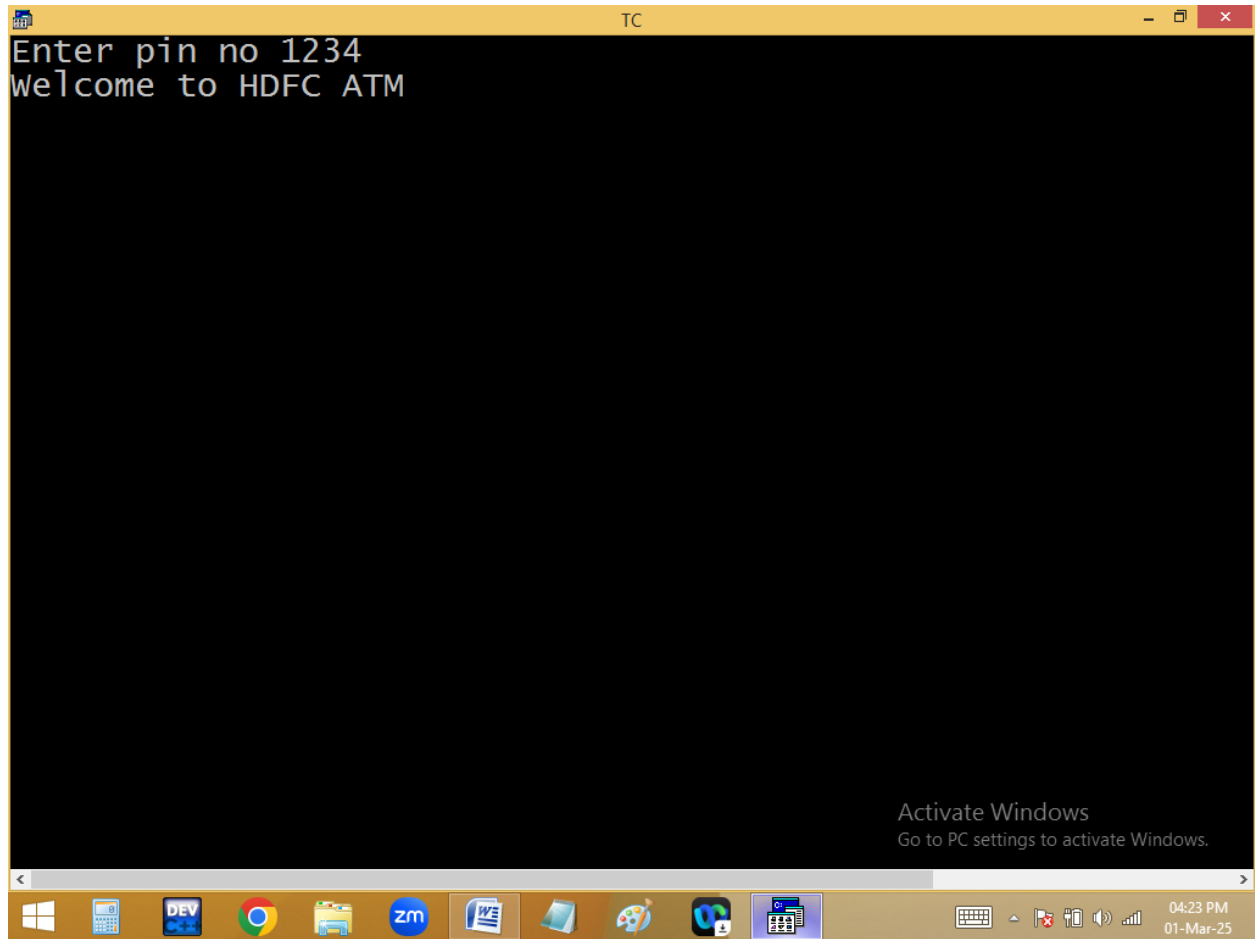
**ATM pin validation:**

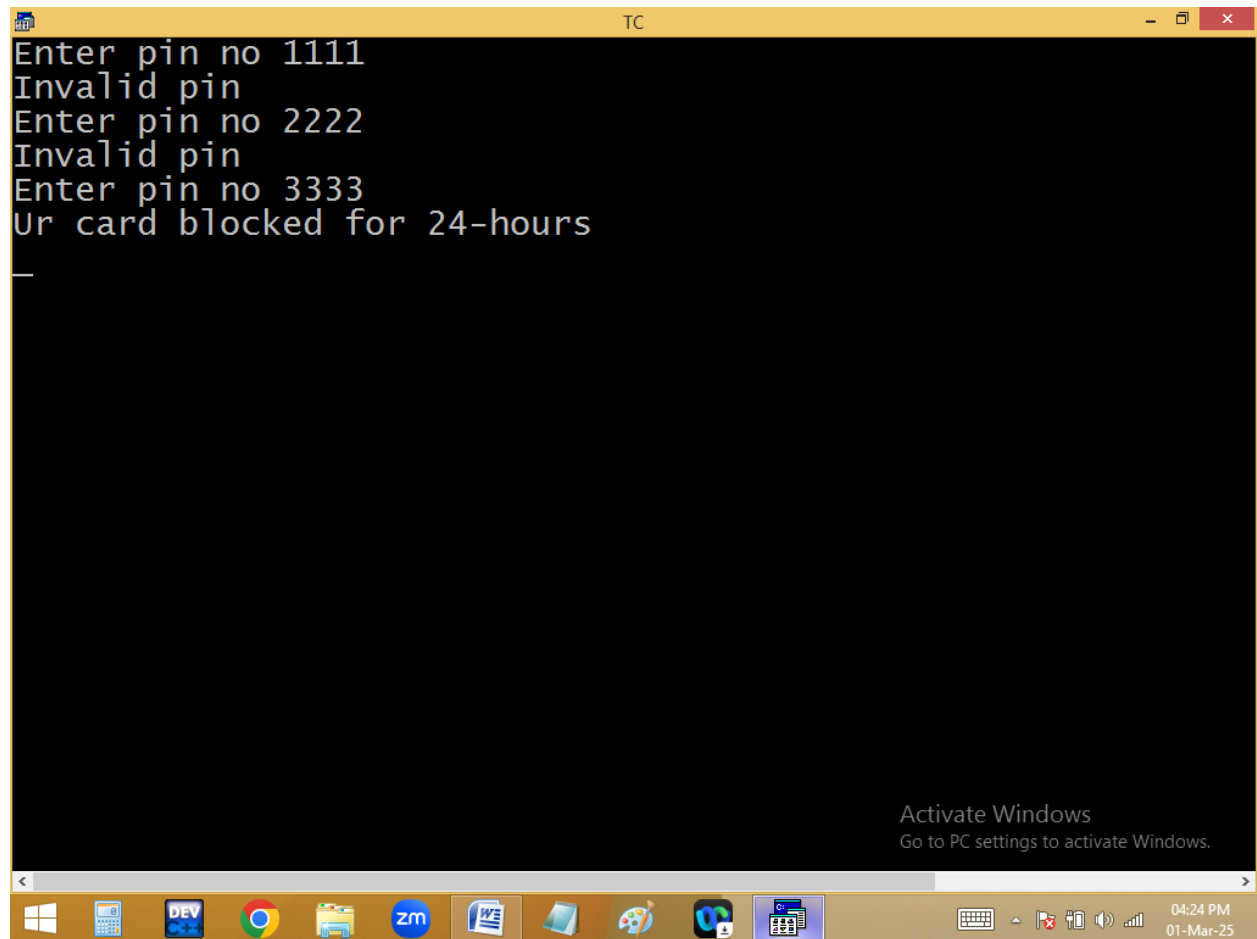


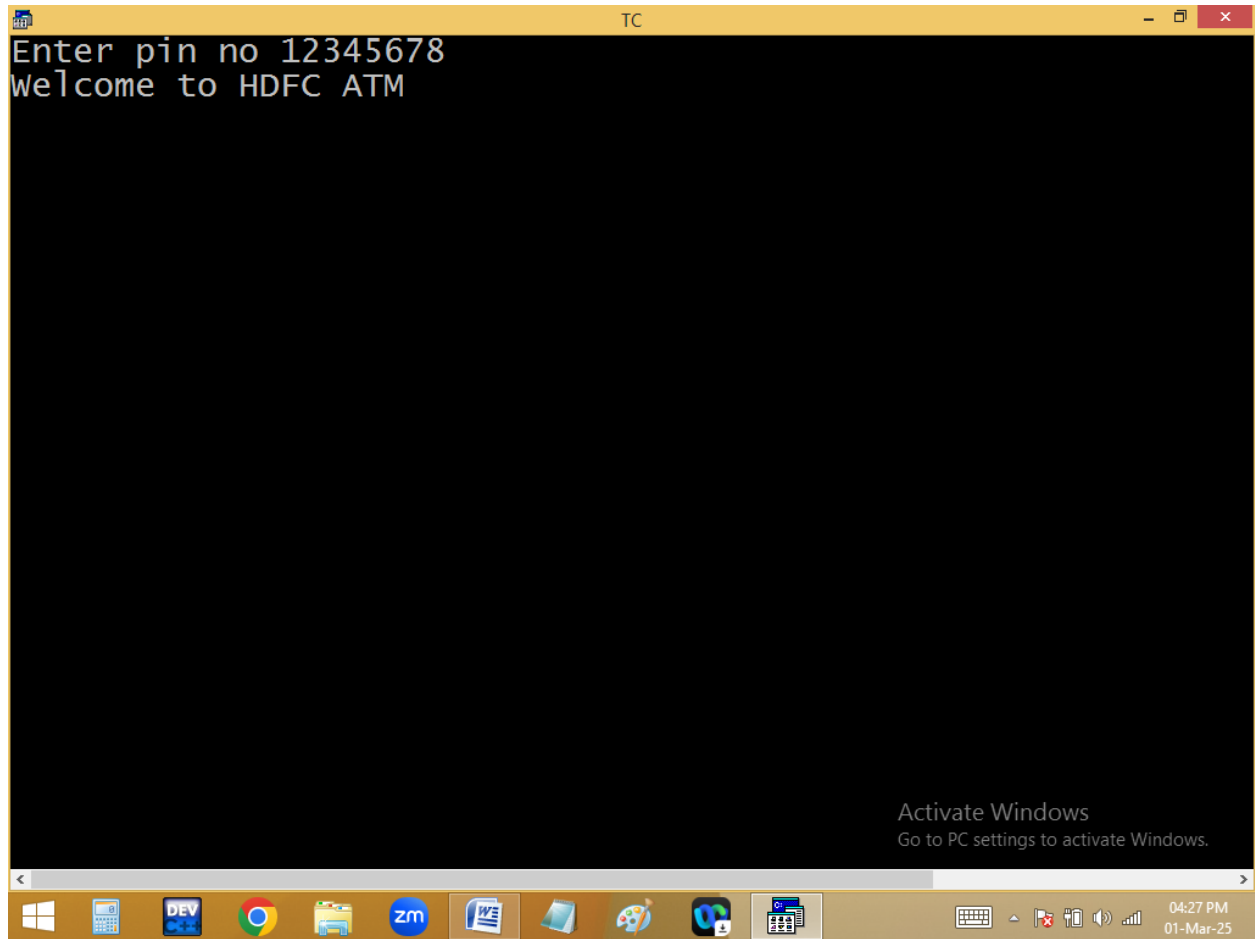
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 1", "Col 2", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "\*", and "E". The main editing area has a blue background with yellow text. It contains a C program for ATM PIN validation. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares `pin` and `count` variables, clears the screen, and enters a loop labeled `abc:`. The loop prompts the user to enter a PIN, checks if it is 1234 (welcome), or if the count reaches 3 (card blocked), or if it is invalid (goto `abc`). The program ends with `getch()`. At the bottom right of the IDE, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock shows "04:23 PM" and "01-Mar-25".

```
#include<stdio.h>
#include<conio.h>
void main()
{
int pin,count=0;
clrscr();
abc:
printf("Enter pin no "); scanf("%4d",&pin);
if(pin==1234)puts("welcome to HDFC ATM");
else
{
count++;
if(count==3)puts("Ur card blocked for 24-hours");
else { puts("Invalid pin"); goto abc;}
}
getch();
}
```

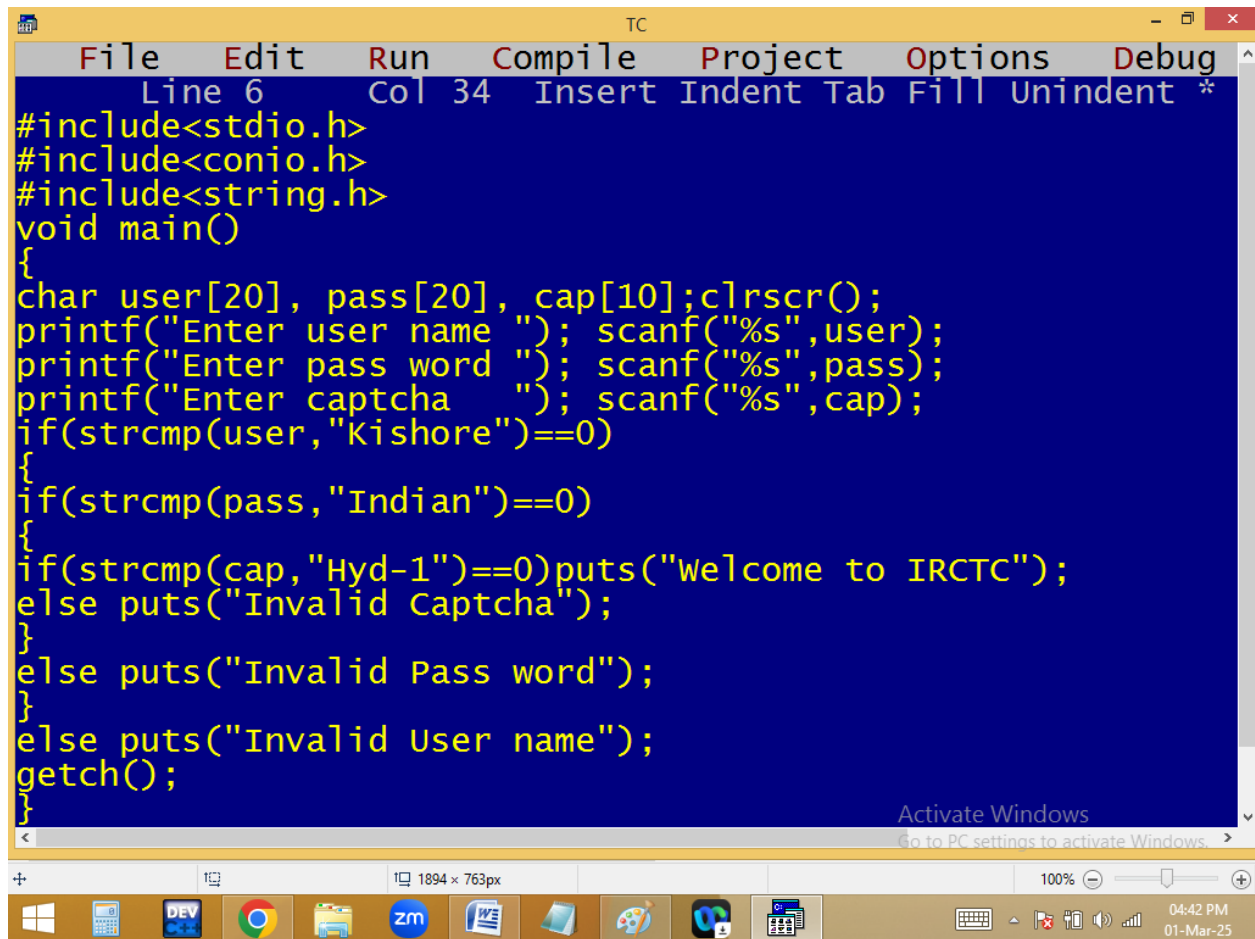
Activate Windows  
Go to PC settings to activate Windows.





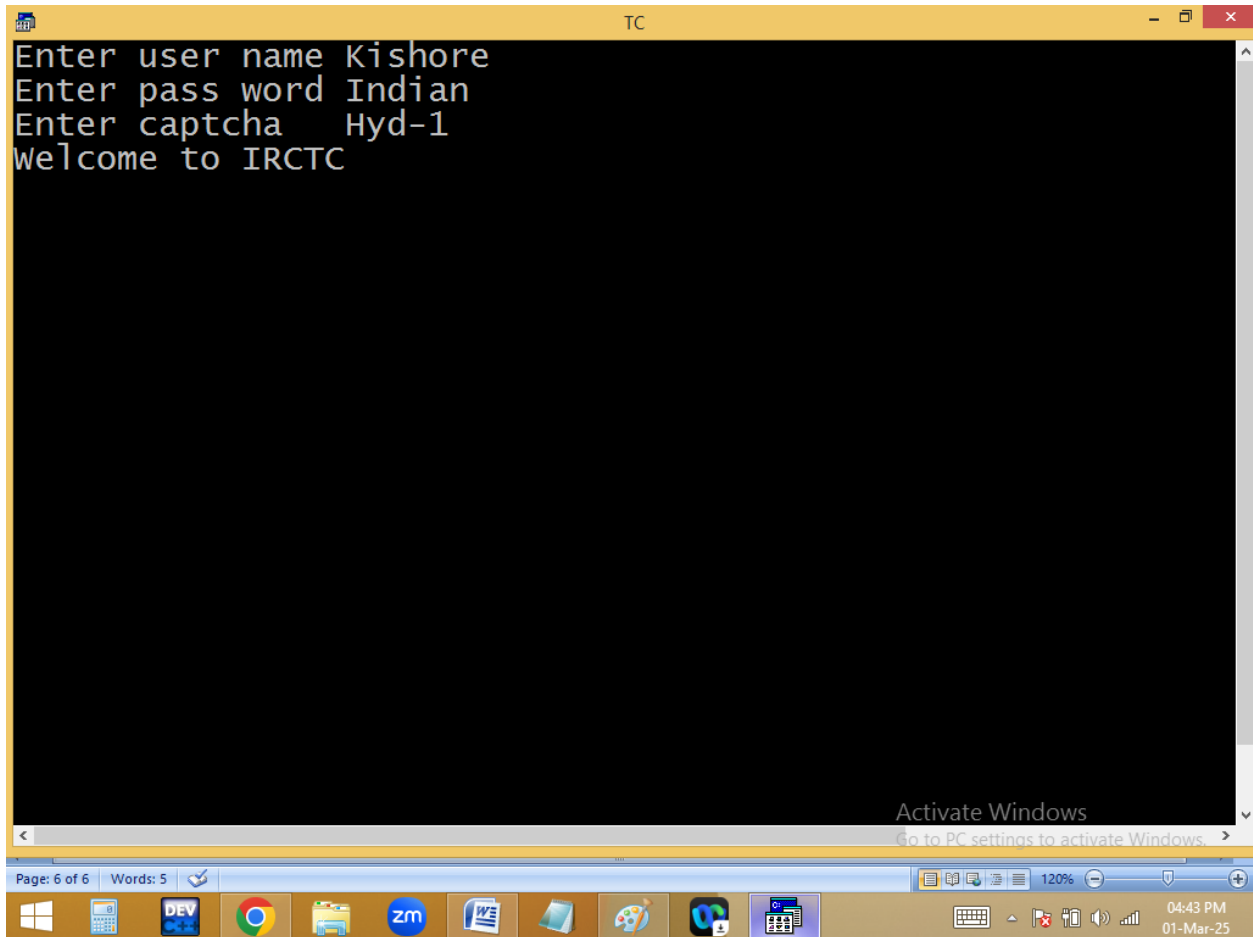


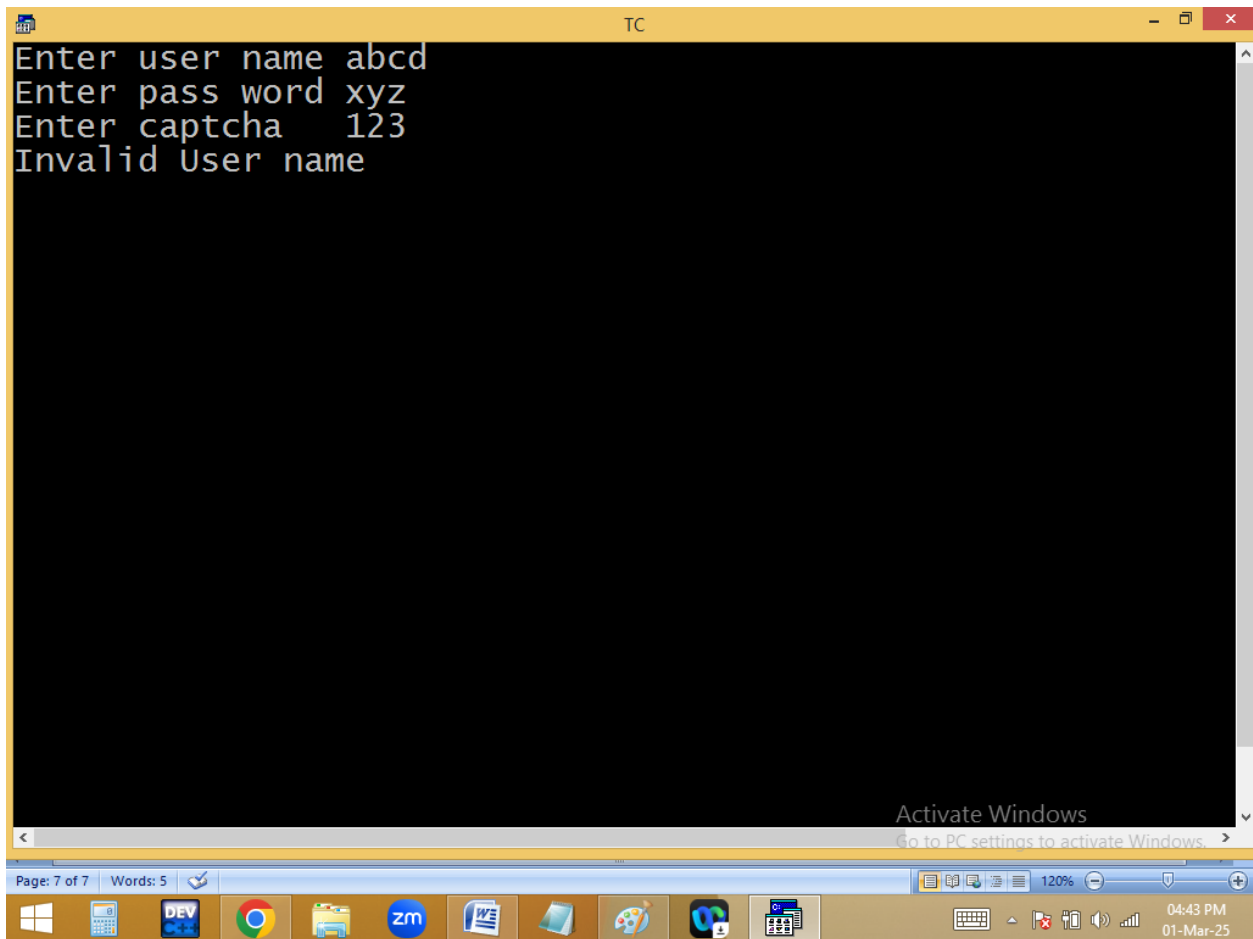
**IRCTC Login:**



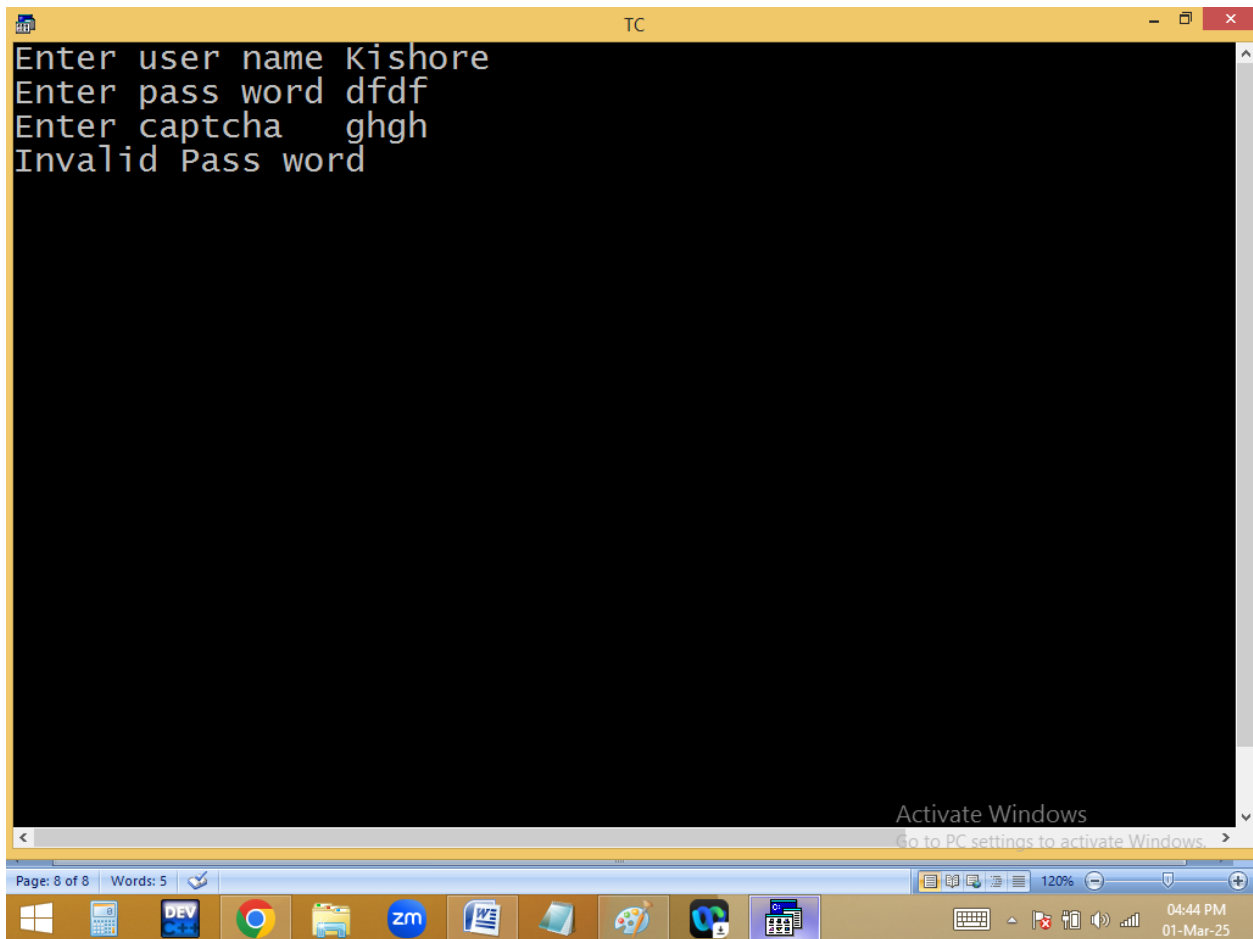
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 34", and provides shortcuts for "Insert", "Indent", "Tab", "Fill", "Unindent", and "\*". The main editing area has a dark blue background with yellow text. It contains a C program for user authentication. The code includes headers for `stdio.h`, `conio.h`, and `string.h`. The `main` function declares arrays for `user`, `pass`, and `cap`, clears the screen, and prompts the user for their name, password, and captcha. It then checks if the entered values match the predefined ones ("Kishore", "Indian", "Hyd-1") and prints appropriate messages: "Welcome to IRCTC" for a successful login, or "Invalid Captcha", "Invalid Pass word", or "Invalid User name" for failures. The program ends with `getch()`. At the bottom right of the IDE, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows". The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock shows "04:42 PM" and "01-Mar-25".

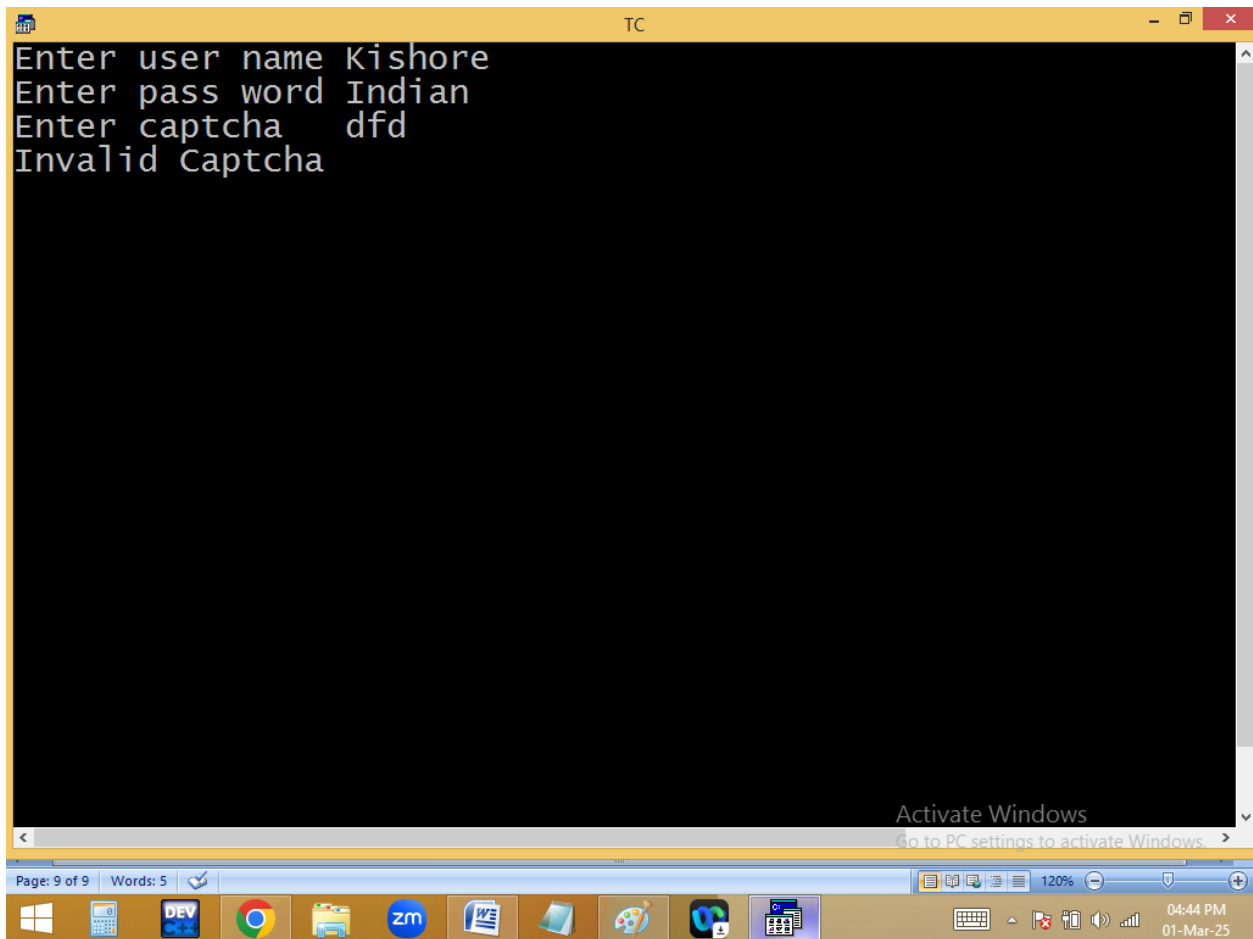
```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[10];clrscr();
printf("Enter user name "); scanf("%s",user);
printf("Enter pass word "); scanf("%s",pass);
printf("Enter captcha  "); scanf("%s",cap);
if(strcmp(user,"Kishore")==0)
{
if(strcmp(pass,"Indian")==0)
{
if(strcmp(cap,"Hyd-1")==0)puts("Welcome to IRCTC");
else puts("Invalid Captcha");
}
else puts("Invalid Pass word");
}
else puts("Invalid User name");
getch();
}
```

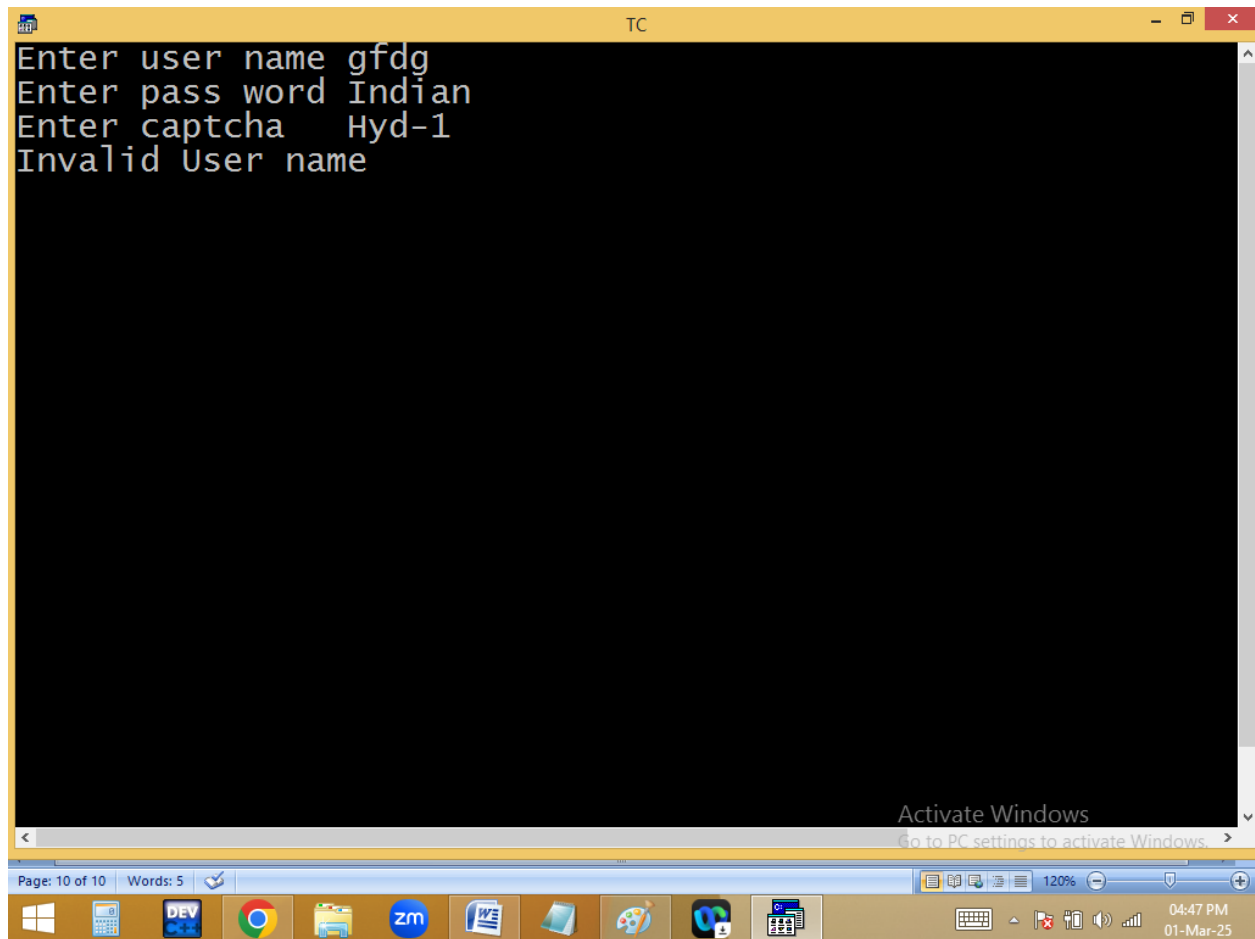




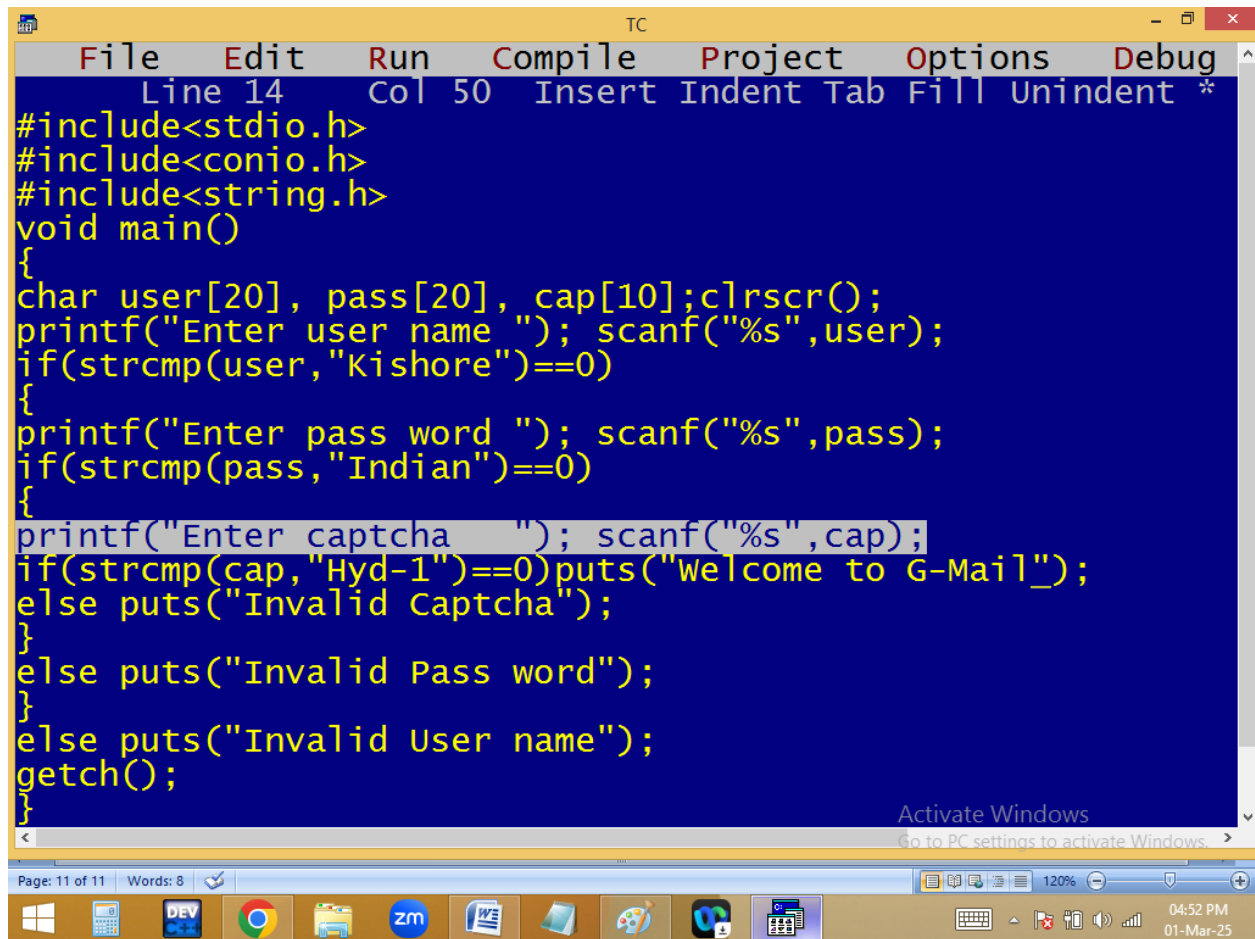








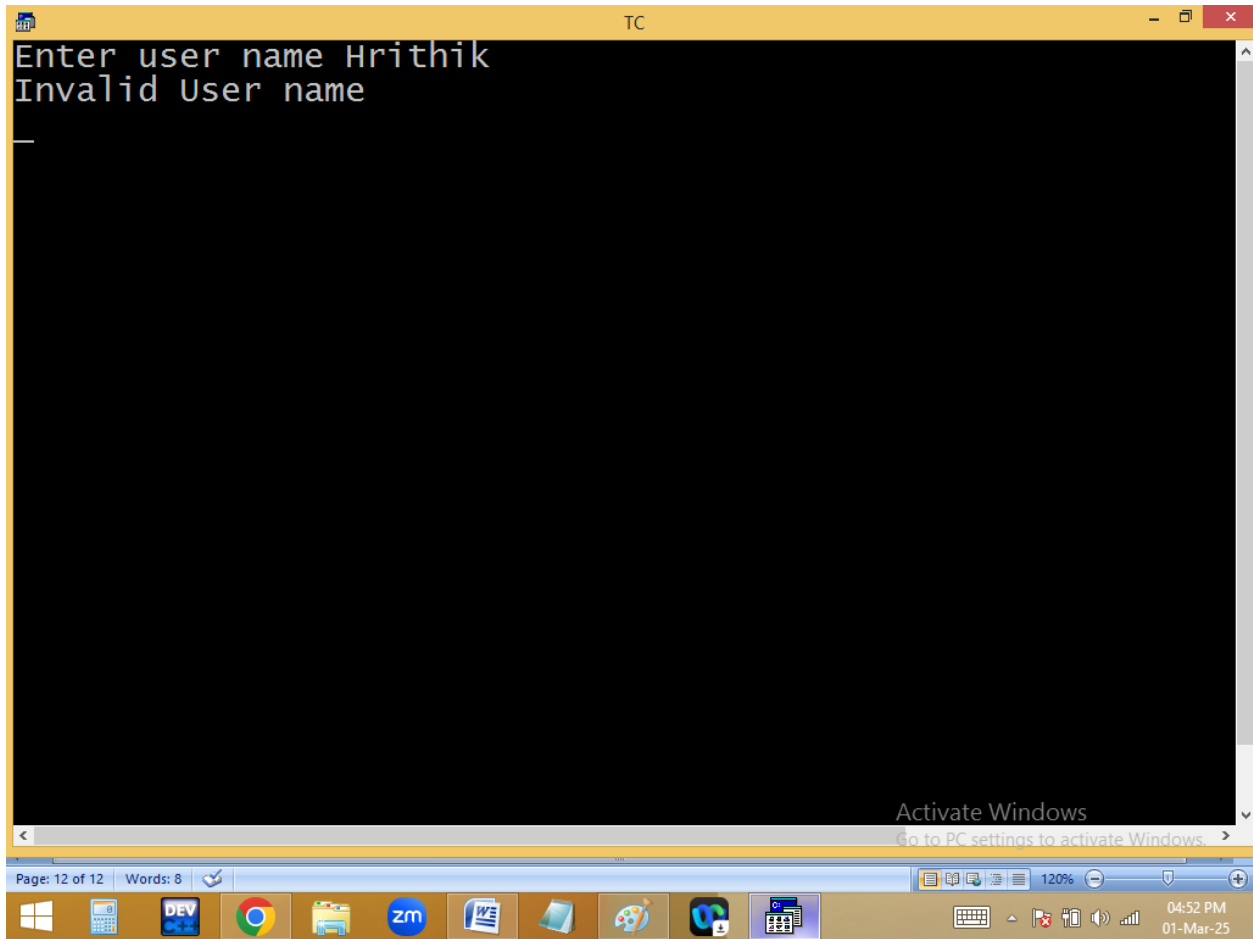
**Gmail style login:**

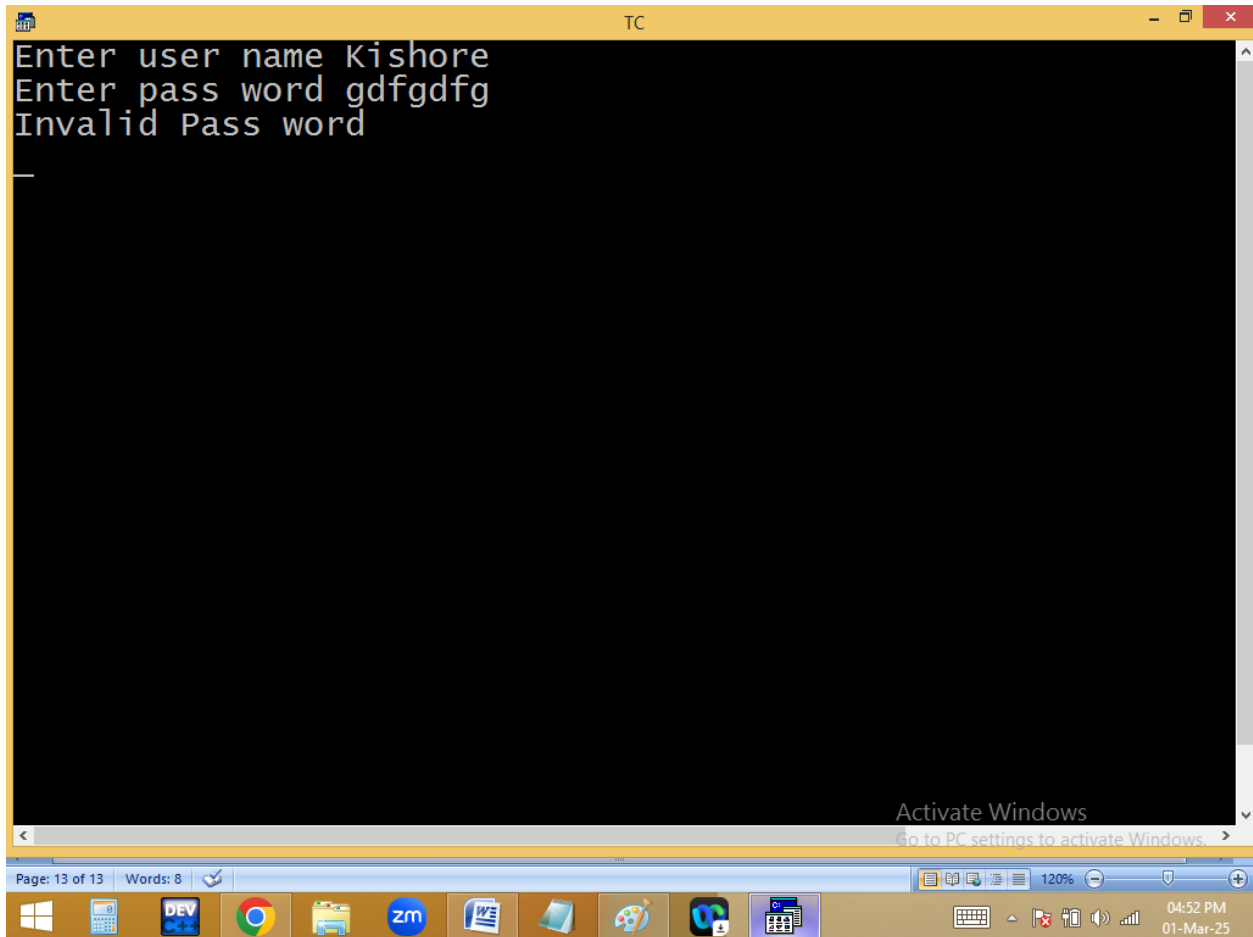


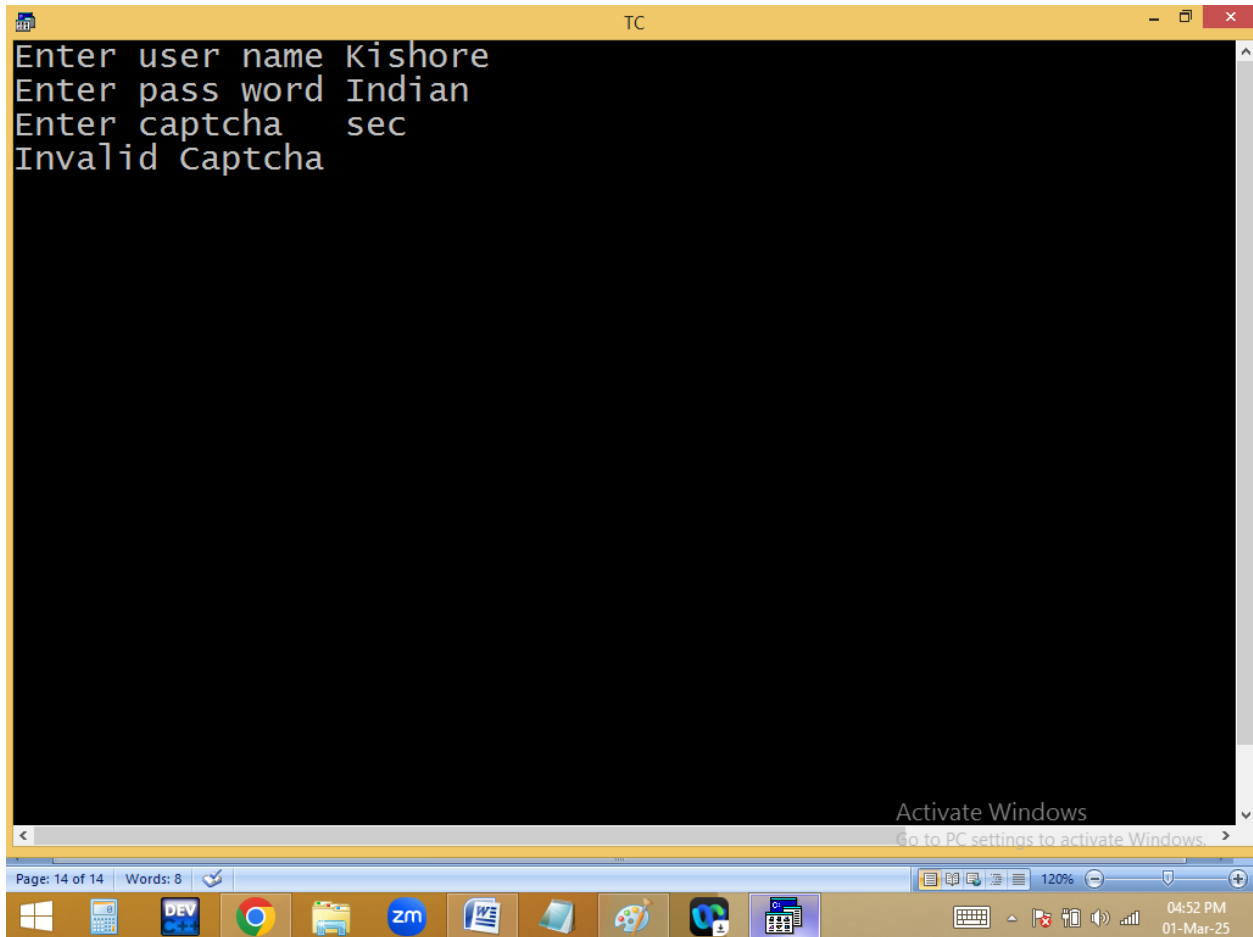
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 50", and "Insert" mode. The code is written in C and is as follows:

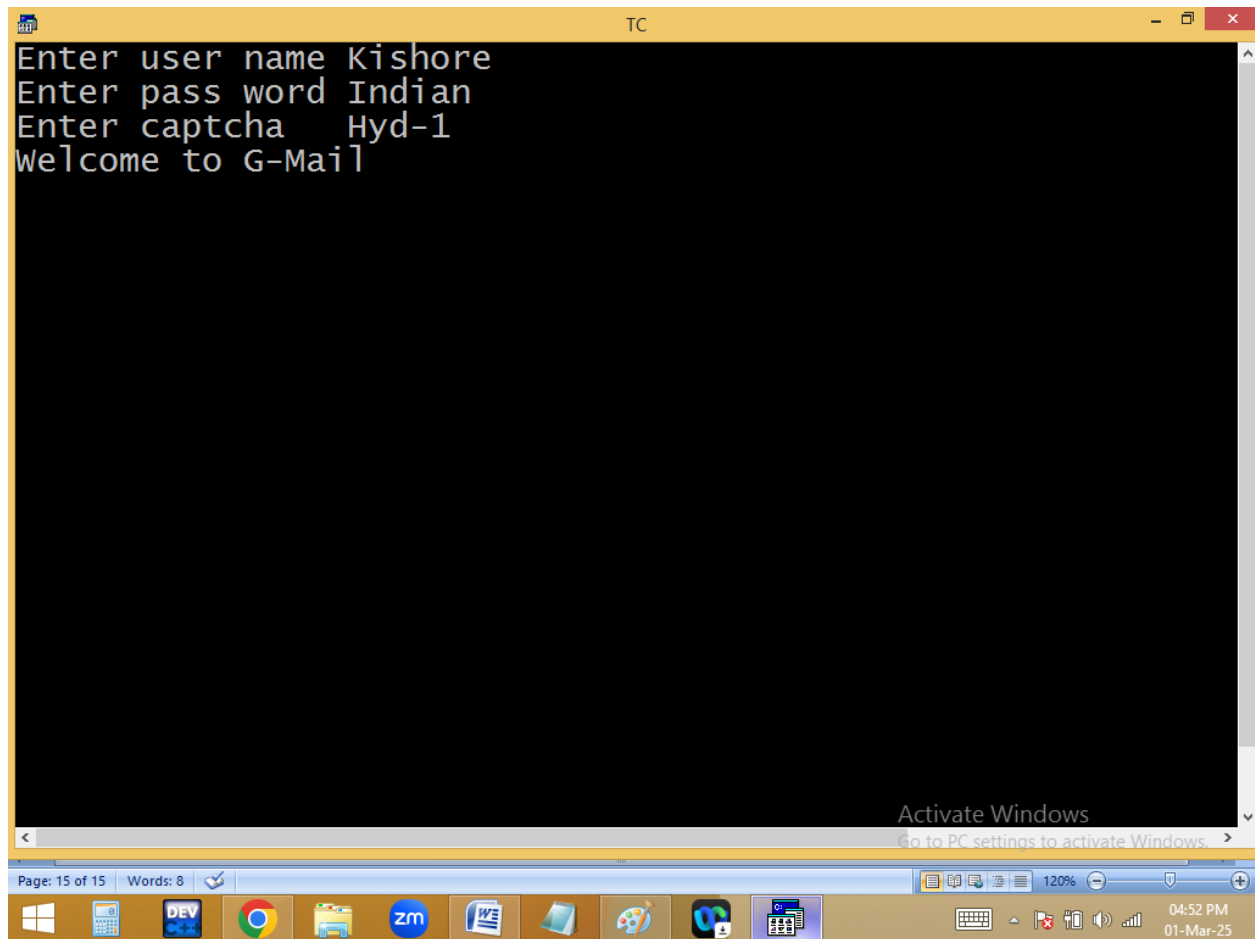
```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[10];clrscr();
printf("Enter user name "); scanf("%s",user);
if(strcmp(user,"Kishore")==0)
{
printf("Enter pass word "); scanf("%s",pass);
if(strcmp(pass,"Indian")==0)
{
printf("Enter captcha "); scanf("%s",cap);
if(strcmp(cap,"Hyd-1")==0)puts("Welcome to G-Mail");
else puts("Invalid Captcha");
}
else puts("Invalid Pass word");
}
else puts("Invalid User name");
getch();
}
```

The bottom status bar shows "Page: 11 of 11" and "Words: 8". The Windows taskbar at the bottom includes icons for Windows, File Explorer, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock shows "04:52 PM" and "01-Mar-25".



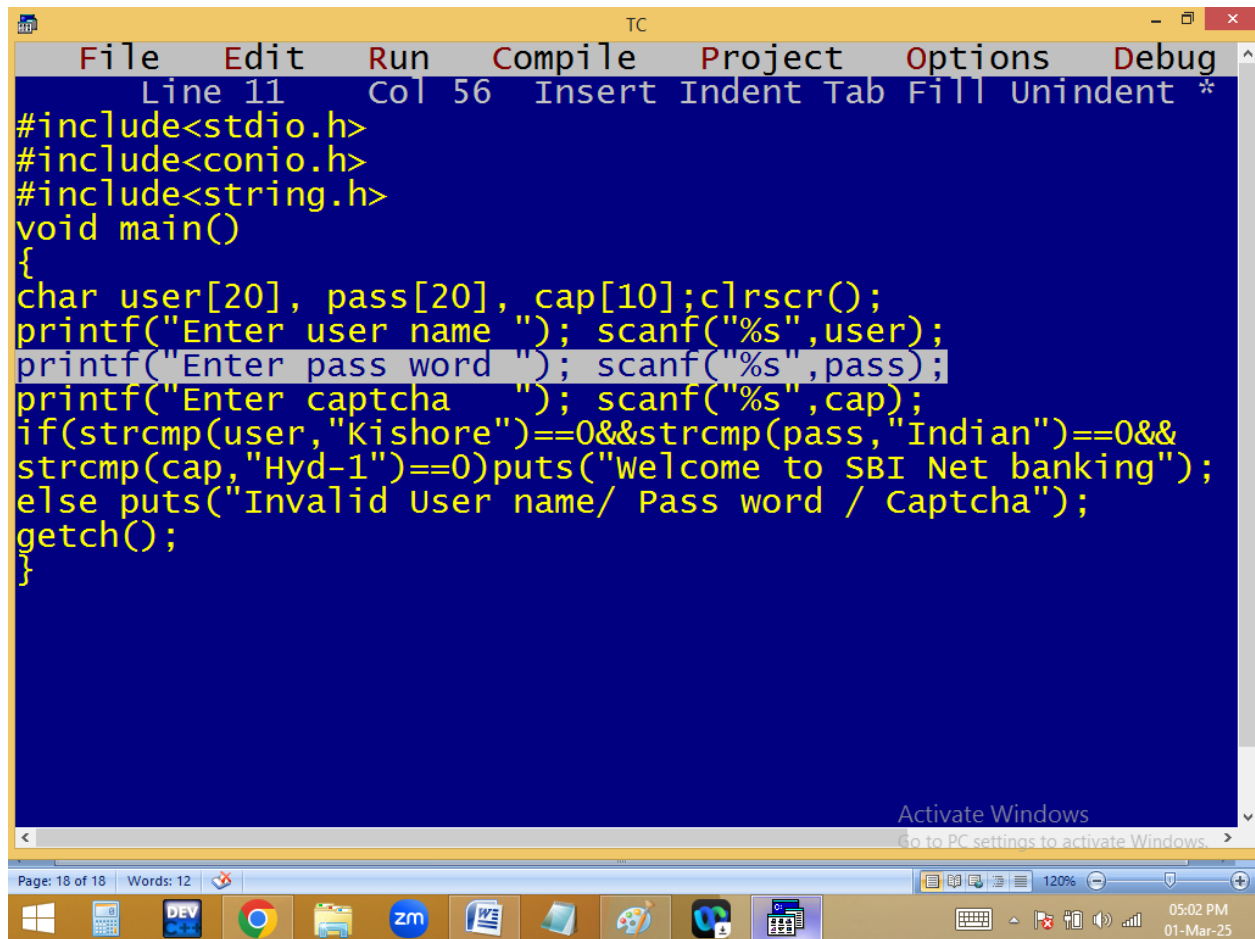






**Internet Banking style login:**

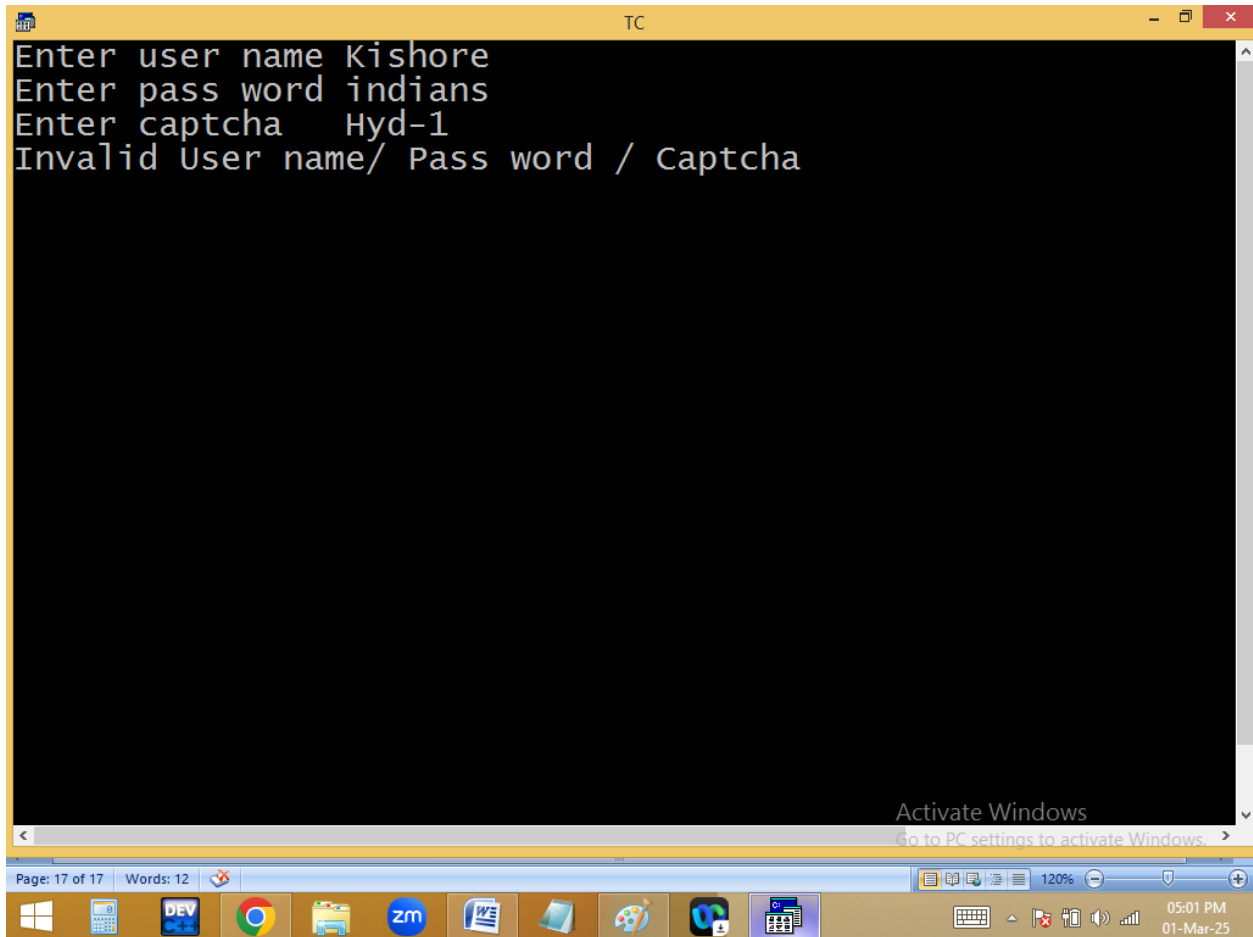




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 11", "Col 56", and "Insert Indent Tab Fill Unindent \*". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[10];clrscr();
printf("Enter user name "); scanf("%s",user);
printf("Enter pass word "); scanf("%s",pass);
printf("Enter captcha "); scanf("%s",cap);
if(strcmp(user,"kishore")==0&&strcmp(pass,"Indian")==0&&
strcmp(cap,"Hyd-1")==0)puts("Welcome to SBI Net banking");
else puts("Invalid User name/ Pass word / Captcha");
getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 18 of 18" and "Words: 12". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the date and time as "05:02 PM 01-Mar-25".



```
TC
Enter user name Kishore
Enter pass word Indian
Enter captcha hyd
Invalid User name/ Pass word / Captcha
```

Activate Windows  
Go to PC settings to activate Windows.

Page: 18 of 19 | Words: 12 | 120%

05:03 PM  
01-Mar-25

Read a stu id, name, 6 sub marks. Find total, avg, grade using below table.

| Avg       | Grade                 |
|-----------|-----------------------|
| $\geq 75$ | Distinction           |
| $\geq 60$ | 1 <sup>st</sup> class |
| $\geq 50$ | 2 <sup>nd</sup> class |
| $\geq 35$ | 3 <sup>rd</sup> class |
| $< 35$    | Fail                  |

```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 35 Insert Indent Tab Fill Unindent *
#include<stdio.h>#include<conio.h>_
void main()
{
int id, tel, eng, hin, mat, sci, soc, tot;char name[30];
float avg; clrscr();
printf("Enter stu id, name, 6 sub marks ");
scanf("%d%s%d%d%d%d%d",&id,name,&tel,&eng,&hin,&mat,&sci,
tot=tel+eng+hin+mat+sci+soc; avg=tot/6.0;
puts("Id\tName\tTot\tAvg\tGrade");
puts("-----");
printf("%d\t%s\t%d\t%.2f\t",id,name,tot,avg);
if(tel>=35&&eng>=35&&hin>=35&&mat>=35&&sci>=35&&soc>=35)
{
if(avg>=75)puts("Distinction");
else if(avg>=60)puts("1st class");
else if(avg>=50)puts("2nd class");
else puts("3rd class");
}
else puts("Failed");
getch();
}
```

Activate Windows  
Go to PC settings to activate Windows.

Page: 20 of 21 Words: 2/42 120% 05:18 PM 01-Mar-25

TC

Enter stu id, name, 6 sub marks 1 vamsi 67 87 90 89 98 87

| Id | Name  | Tot | Avg   | Grade       |
|----|-------|-----|-------|-------------|
| 1  | vamsi | 518 | 86.33 | Distinction |

Activate Windows  
Go to PC settings to activate Windows.

Page: 21 of 21 | Words: 42

120%

05:18 PM  
01-Mar-25

```
TC
Enter stu id, name, 6 sub marks 2 kumar 45 40 55 45 56 54
Id      Name      Tot      Avg      Grade
-----
2      kumar      295      49.17     3rd class

```

Activate Windows  
Go to PC settings to activate Windows.

Page: 22 of 22 | Words: 42 | 120%

05:18 PM  
01-Mar-25

```
TC
Enter stu id, name, 6 sub marks 3 vijay 44 22 35 45 54 40
Id      Name      Tot      Avg      Grade
-----
3      vijay      240      40.00    Failed

```

Activate Windows  
Go to PC settings to activate Windows.

Page: 23 of 23 | Words: 42 | 120%

05:18 PM  
01-Mar-25

**Eg: Currency Denomination: 333Rs**

**500-0**

**200-1**

**100-1**

**50-0**

**20-1**

**10-1**

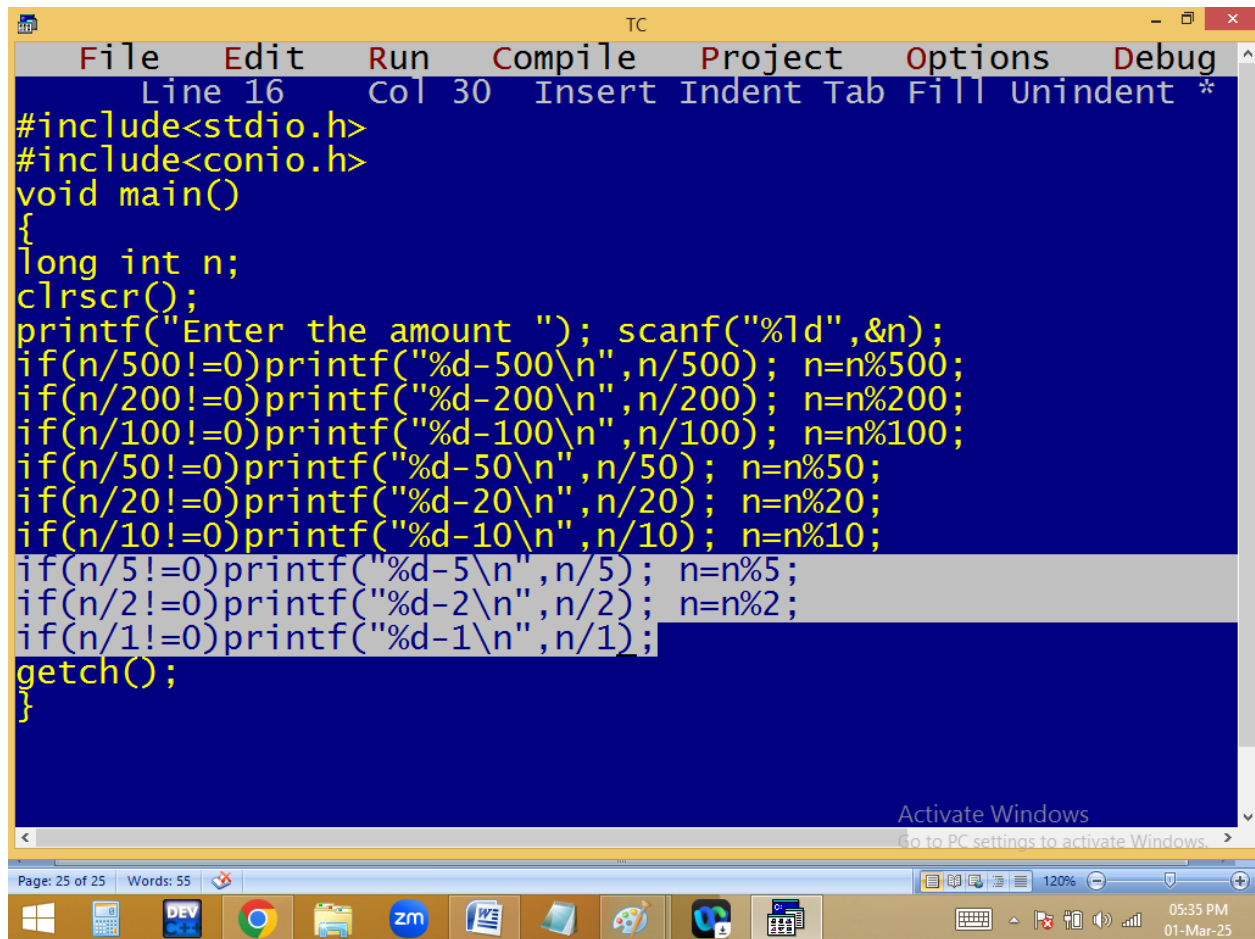
**5-0**

2-1

1-1

$$\begin{array}{l} 500 \overline{) 333} \quad (0-500 \\ 200 \overline{) 333} \quad (1-200 \\ 100 \overline{) 133} \quad (1-100 \\ 50 \overline{) 33} \quad (0-50 \\ 20 \overline{) 33} \quad (1-10 \\ \quad \quad \quad \frac{10}{13} \end{array} \quad \begin{array}{l} 10 \overline{) 13} \quad (1-10 \\ \quad \quad \frac{10}{3} \\ 2 \overline{) 3} \quad (5-0 \\ \quad \quad \frac{2}{1} \quad 1=2 \\ \quad \quad \frac{1}{1-1} \end{array}$$

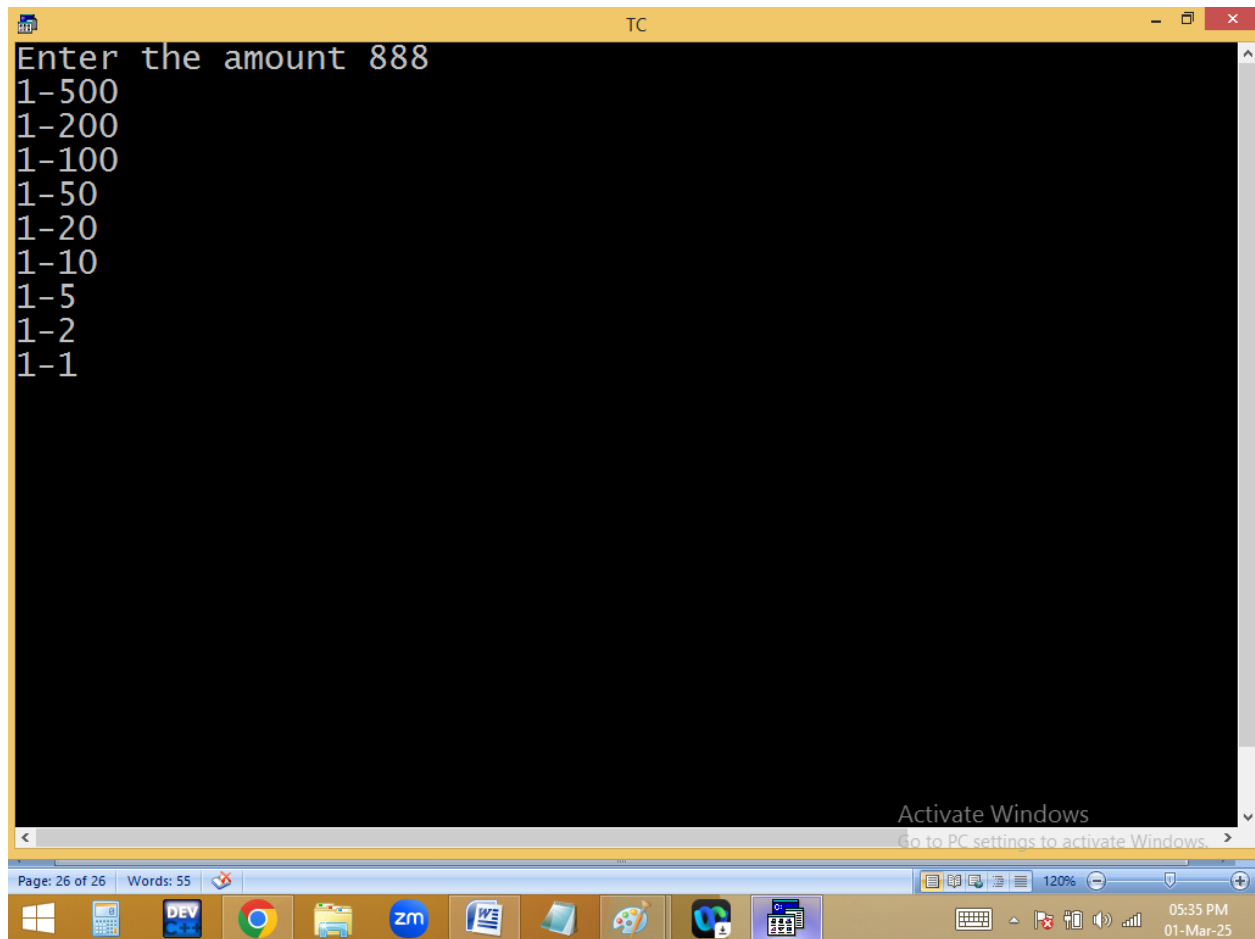




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 16", "Col 30", and "Insert" mode. The code is written in C and is designed to find the prime factors of a number entered by the user. The code uses a series of if statements to check for divisibility by 500, 200, 100, 50, 20, 10, 5, 2, and 1, printing the factor and the remainder each time. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int n;
    clrscr();
    printf("Enter the amount "); scanf("%ld",&n);
    if(n/500!=0)printf("%d-500\n",n/500); n=n%500;
    if(n/200!=0)printf("%d-200\n",n/200); n=n%200;
    if(n/100!=0)printf("%d-100\n",n/100); n=n%100;
    if(n/50!=0)printf("%d-50\n",n/50); n=n%50;
    if(n/20!=0)printf("%d-20\n",n/20); n=n%20;
    if(n/10!=0)printf("%d-10\n",n/10); n=n%10;
    if(n/5!=0)printf("%d-5\n",n/5); n=n%5;
    if(n/2!=0)printf("%d-2\n",n/2); n=n%2;
    if(n/1!=0)printf("%d-1\n",n/1);
    getch();
}
```

The IDE interface includes a status bar at the bottom showing "Page: 25 of 25" and "Words: 55". The Windows taskbar is visible at the bottom, showing the Start button, taskbar icons for DEV, Chrome, File Explorer, Zoom, Word, and other applications, and the system tray with the date and time "05:35 PM 01-Mar-25".



The screenshot shows a Turbo C++ (TC) compiler window with a black background and white text. The text displays the output of a program: "Enter the amount 888" followed by a list of numbers from "1-500" down to "1-1". The window title bar says "TC". At the bottom of the window, there is a status bar with "Page: 26 of 26" and "Words: 55". The Windows taskbar is visible at the bottom, showing various icons and the system clock indicating 05:35 PM on 01-Mar-25.

```
Enter the amount 888
1-500
1-200
1-100
1-50
1-20
1-10
1-5
1-2
1-1
```

## Ternary / Conditional operator( ? : )

It is similar to if else / ladder if in working style.

It allows to complete if else / ladder if in a single statement.

When we are working with if else/ladder if it is going to take more than one line of

statements. Ternary operator is going to finish the same task in a single statement.

**But the difference between if ...else and ternary operator is ternary operator supports only one statement at a time and if supports any number of statements.**

It is having 3 expressions. Hence it is called ternary operator.

It is starting with a condition. Hence it is called conditional operator.

**Syntax:**

condition ? true statement : false statement ;

↑                    ↑                    ↑

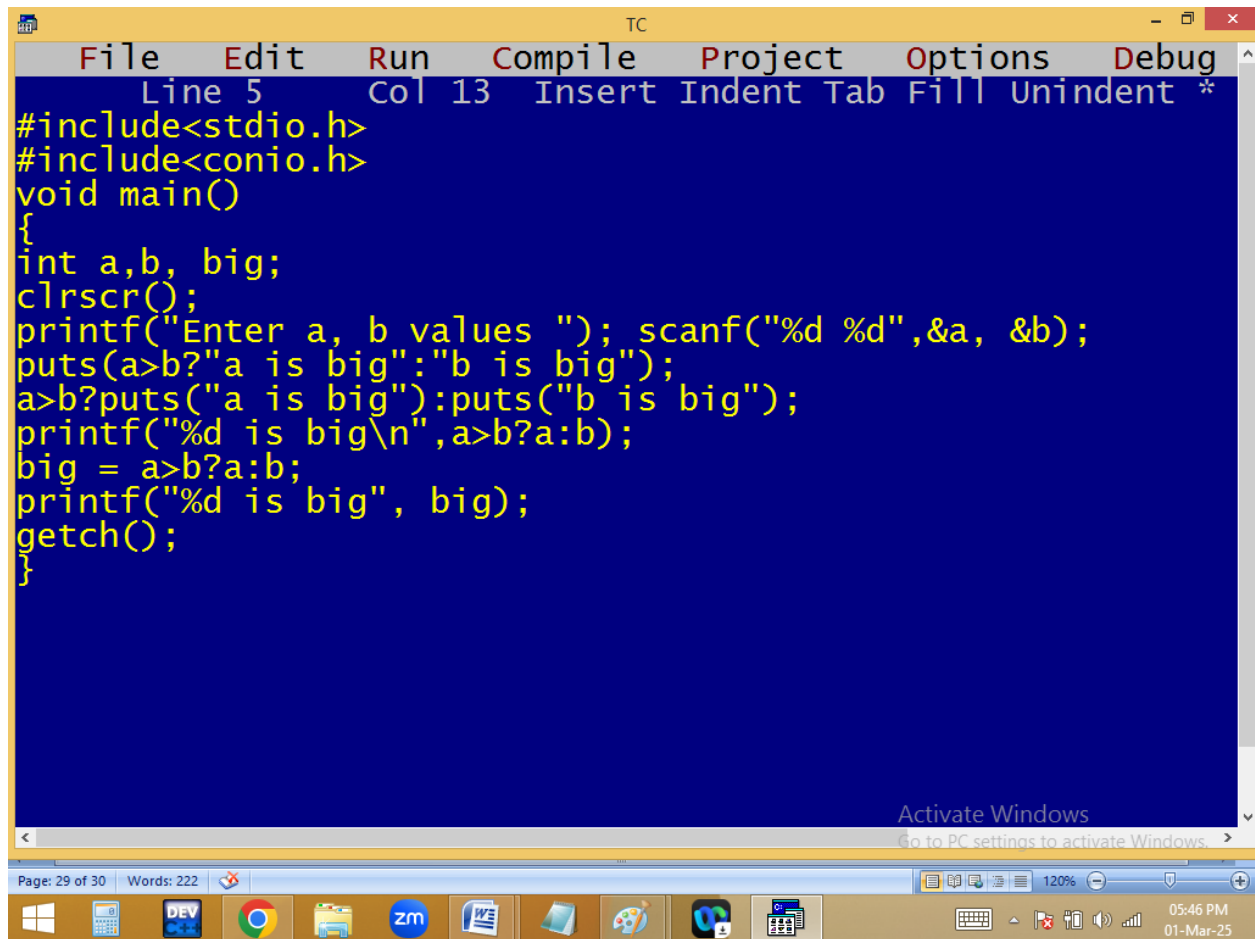
exp1/op1      exp2/op2      exp3/op3

If condition true, statement after **?** executed.

If condition false, statement after **:** is executed.

When compared with if else, conditional operator **performance is high**.

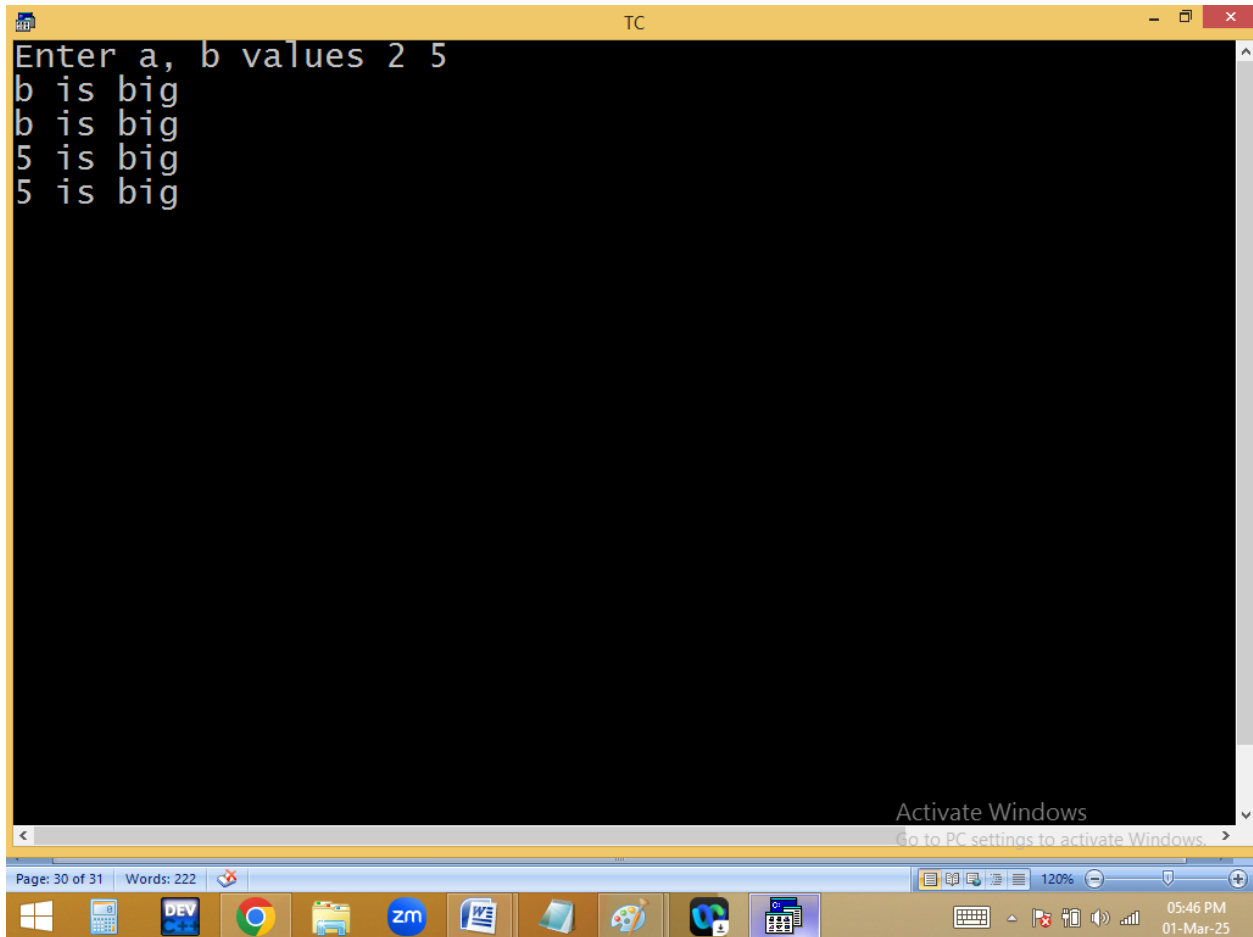
**Eg:** Finding big in two no's using ternary operator.

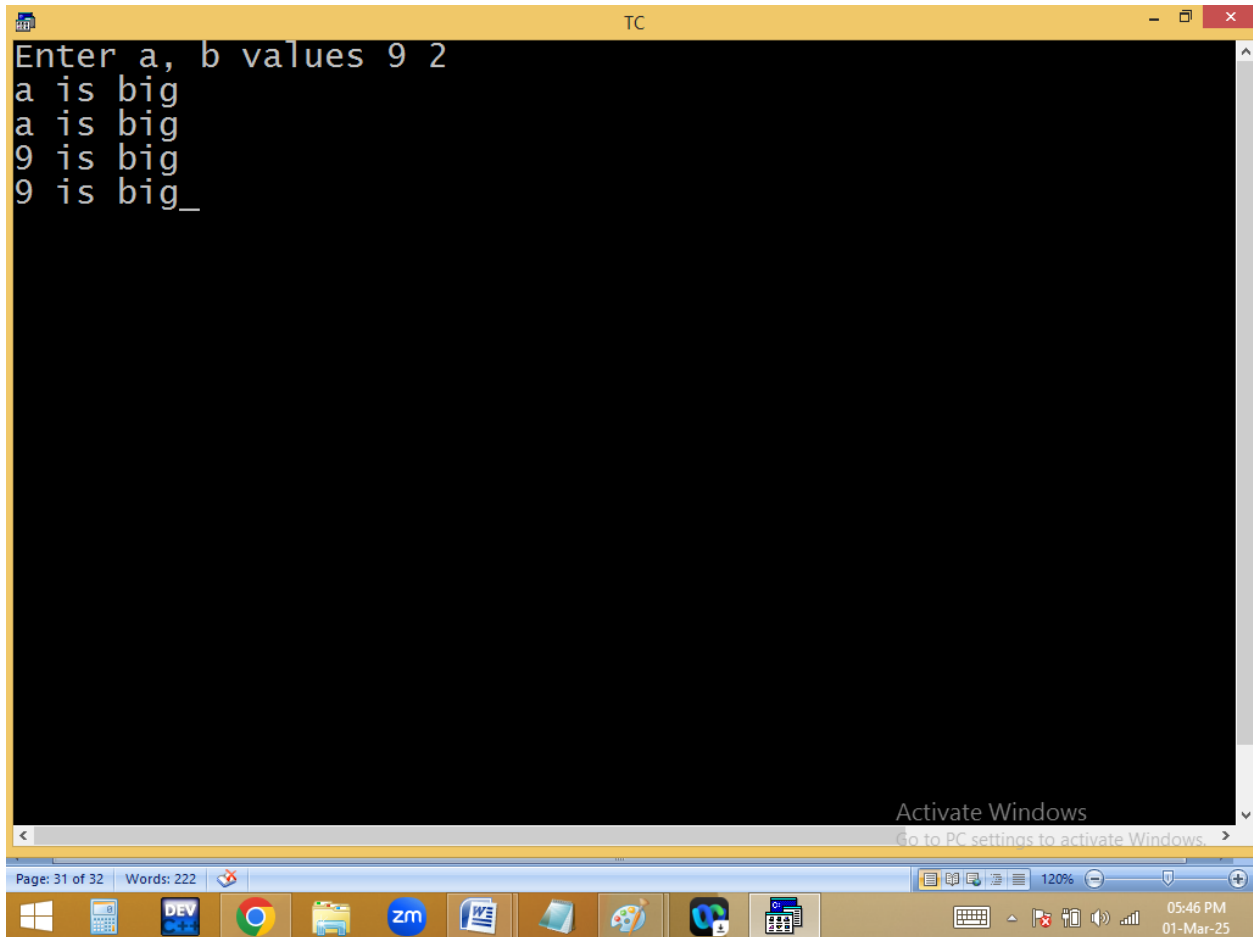


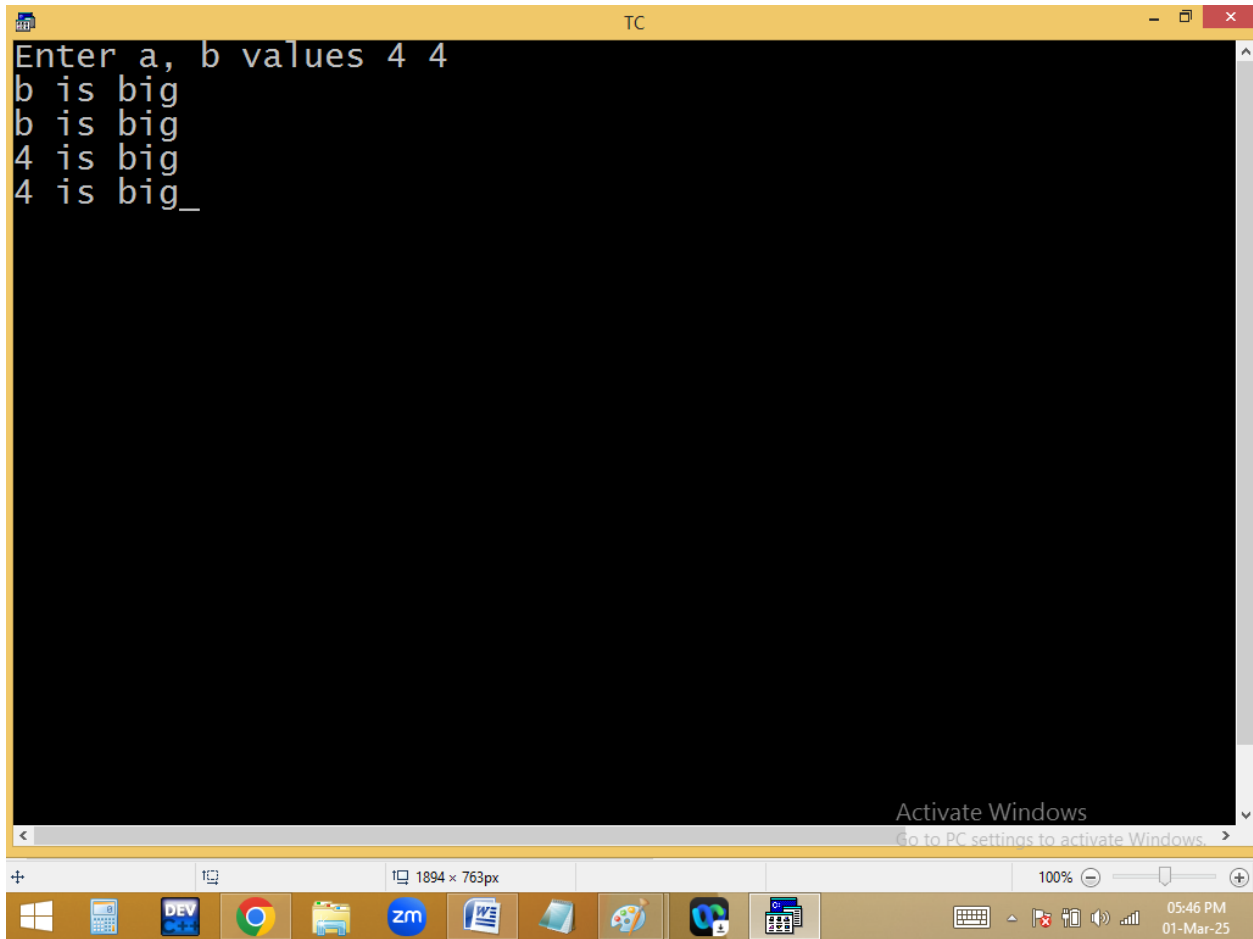
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 5", "Col 13", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and an asterisk. The main editing area has a dark blue background with yellow text. It contains a C program that takes two integers as input and prints the larger one. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b, big;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
puts(a>b?"a is big":"b is big");
a>b?puts("a is big"):puts("b is big");
printf("%d is big\n",a>b?a:b);
big = a>b?a:b;
printf("%d is big", big);
getch();
}
```

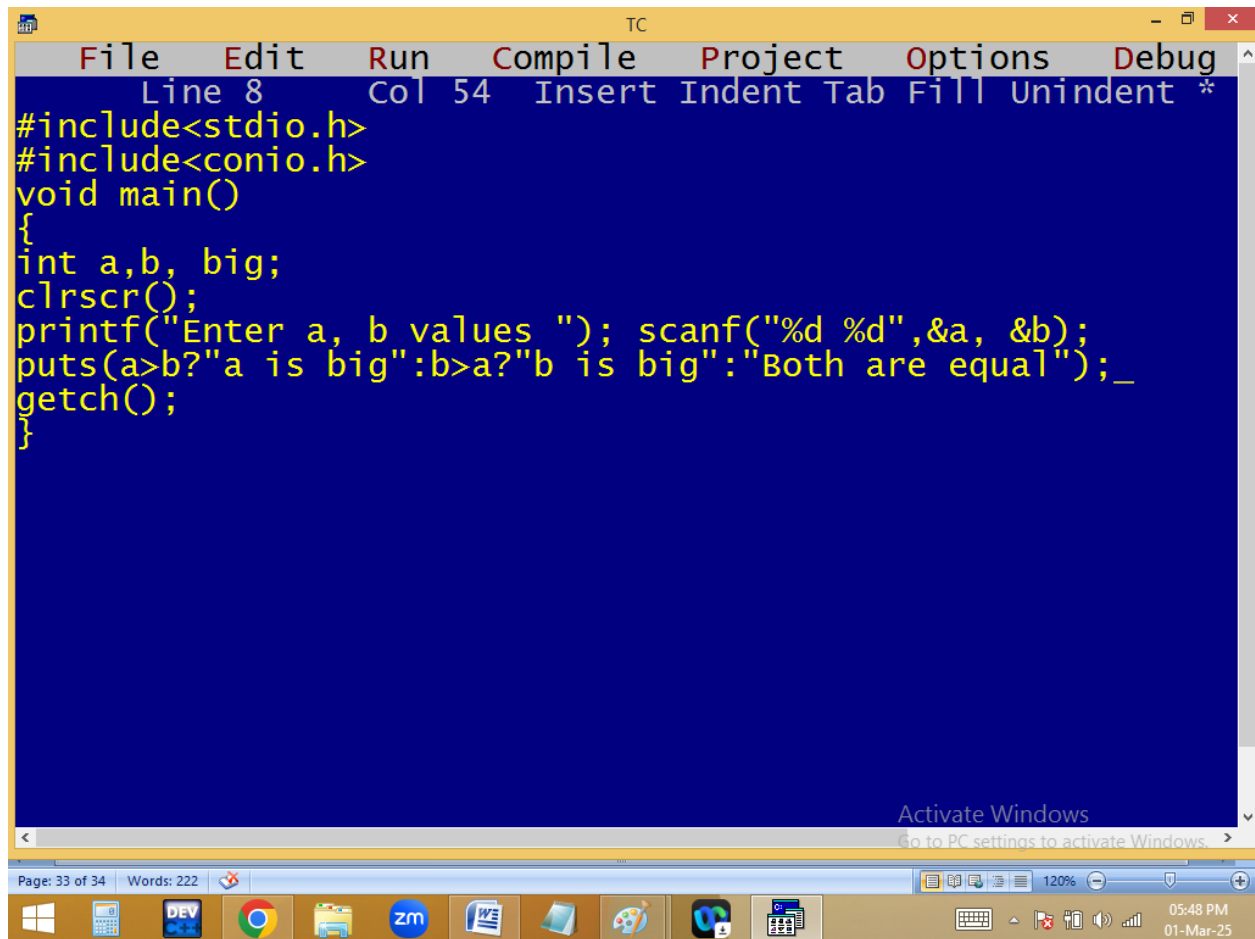
At the bottom of the IDE window, there is a status bar with "Page: 29 of 30" and "Words: 222". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the time as 05:46 PM and the date as 01-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







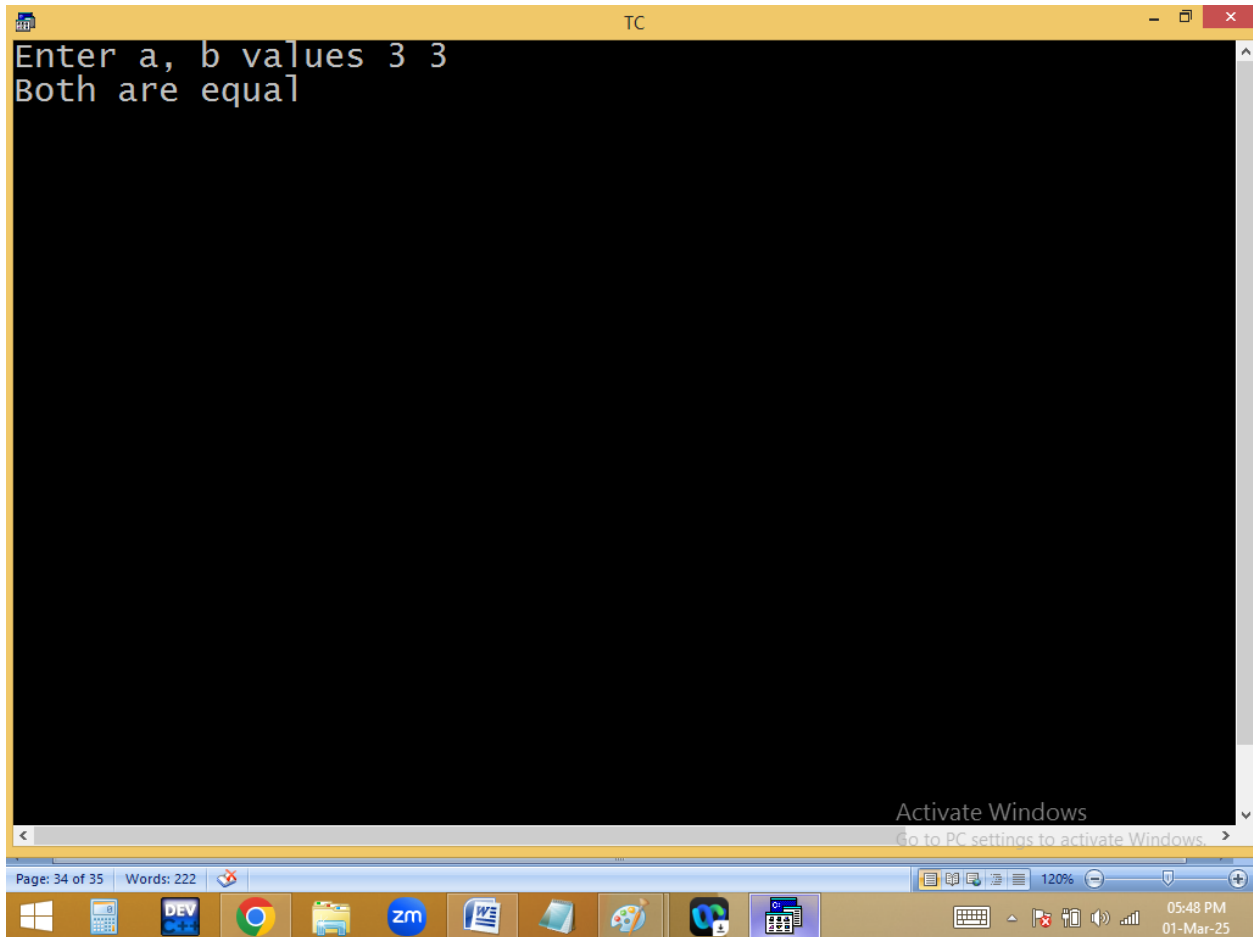


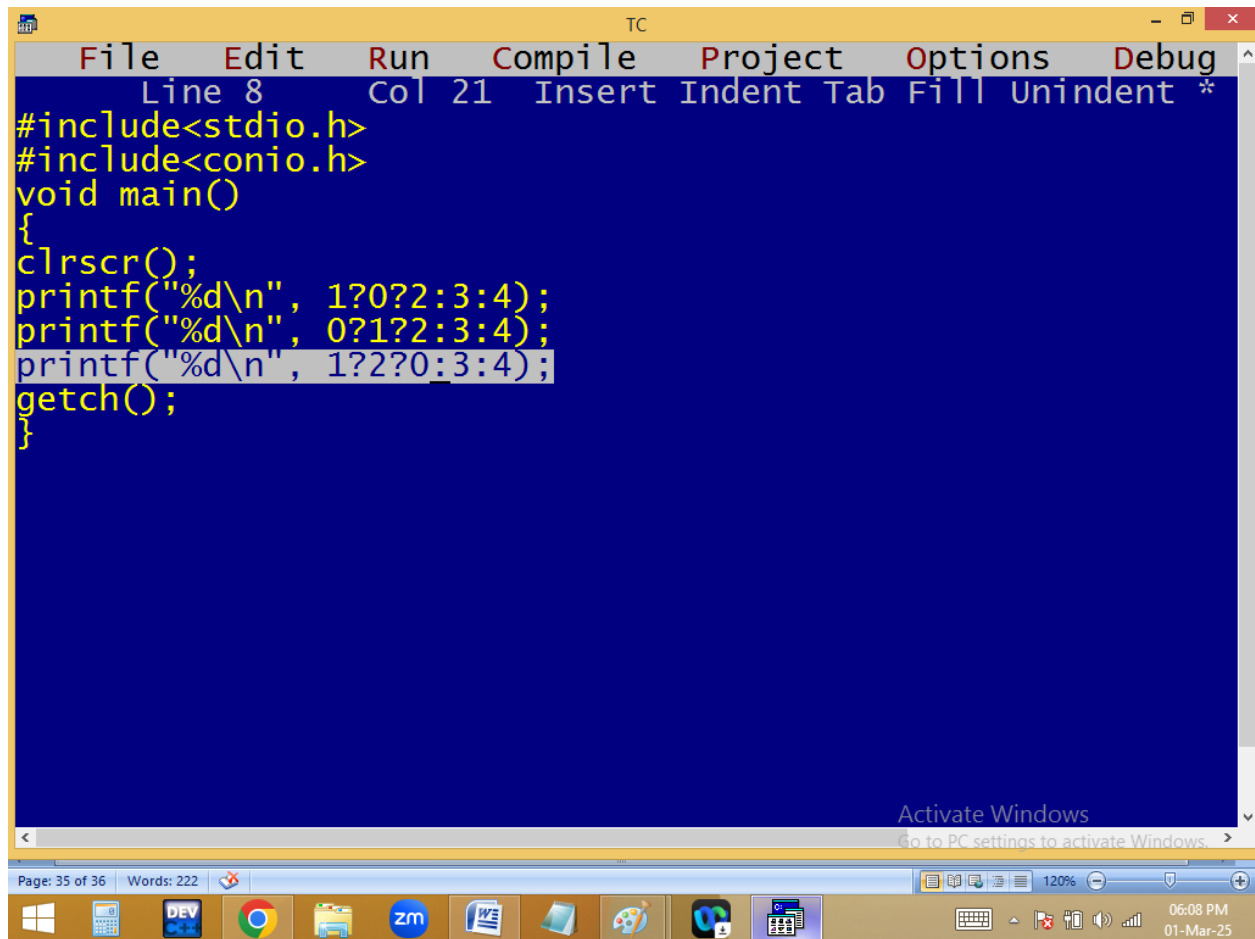


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 8" and "Col 54". The editor area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b, big;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
puts(a>b?"a is big":b>a?"b is big":"Both are equal");_
getch();
}
```

At the bottom of the window, there is a taskbar with various icons including Windows, a calculator, a folder, a ZOOM icon, a document, a folder, a paint palette, a globe, and a calendar. The system tray on the right shows the date and time: "05:48 PM 01-Mar-25".

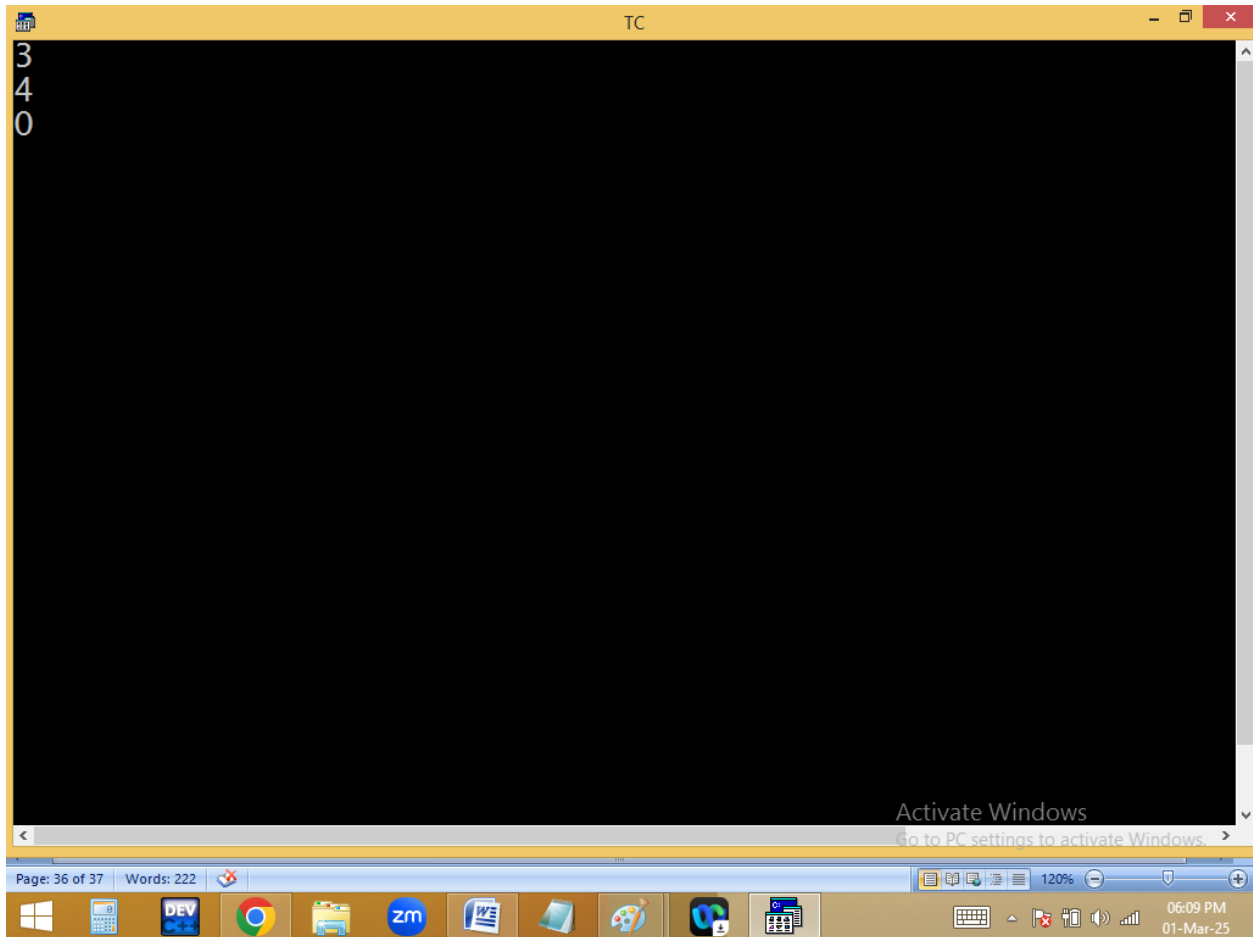




The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 21", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and an asterisk. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 1?0?2:3:4);
printf("%d\n", 0?1?2:3:4);
printf("%d\n", 1?2?0:3:4);
getch();
}
```

The third line of the printf statement is highlighted with a light blue selection bar. At the bottom of the window, there is a status bar with "Page: 35 of 36" and "Words: 222". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Zoom, Word, and other applications. The system tray on the right shows the date and time: "06:08 PM 01-Mar-25".



**Finding even/odd in single statement?**

**`puts(n%2==0?"Even":"Odd");`**

**`if(n%2==0)p(even);  
else p(odd);`**

